To what extent did Artificial Intelligence Revolutionize the Medical Industry

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Abstract

This essay aims to discover the effect of AI in the Medical Industry. This has been done by investigating the application of AI in two distinct ways but are also categorized as Medical Biology. Then the effects it would bring would be analysed based on scientific findings that were conducted before, and opinions from different people or authorities in the field of medical Biology. The conclusion that can be drawn from the investigation is the AI has helped humans in the Medical Industry but cannot replace humans due to the lack of trust. Therefore the application of AI cannot efficiently revolutionize the Medical Industry.

Introduction

After the conference[1] about Artificial Intelligence in 1956 in Carnegie Mellon University (CMU) and Massachusetts Institute of Technology (MIT), AI, initially perceived as 'Almost Impossible', has evolved to something everyone uses in daily life, ranging from Siri to video card decoding even without knowing that they are already using AI.

Nowadays, AI has changed how a lot of fields functions, and one of which is the medical industry because it can do something more accurately and learn based on what was done right and avoid doing what has been done wrong through the process of Machine Learning (ML). It was promised that AI would provide professional suggestions to treatments for 12 cancers contributing to 80% of the world's total cases[2].

However, no matter how AI evolves, humans are still reluctant to trust AI, because they believe the algorithm may go wrong someday and generate an output that cannot be reliable. Therefore, they would rather trust an experienced professional even though if s/he is giving them fraudulent information.[3]

Even though AI has not been trusted completely, this piece of new technology has been adapted in a lot of

[1] **"World-Information.Org".** 2019. World-Information.Org. http://world-information.org/wio/infostructure/100437611663/100438659360.

[2] **Vyacheslav, Polonski.** 2018. The Conversation. http://theconversation.com/people-dont-trust-ai-heres-how-we-can-change-that-87129.

[3] Quoted: 'Yet for all these technological advances, we still seem to deeply lack confidence in AI predictions. Recent cases show that people don't like relying on AI and prefer to trust human experts, even if these experts are wrong.', *ibid*

areas in the Medical world[4]. This essay aims to describe two applications of AI which includes genetic screening and surgical assistance, these topics are chosen because surgeries have been done by surgeons and people share a concern where machines might overtake them in jobs and employment rates, and genetic screening is chosen because it has already been a field where machines are used but humans were used to analysing the data.

Artificial Intelligence: Medicine based on Genome

Genetics has been one of the most recently developed topics. Only requiring 4 DNA bases, a human can be created[5]. Everyone's DNA is different, and there can be 8,324,608[6] combinations of DNA. Despite the fact that over 99% of the DNA is the same across the global population, a right DNA can determine a lot about a person, ranging from food allegories to genetic disorders such as Down Syndrome.

Since DNA is unique, medicine should not be applied on the 'one for all' basis but personally according to the genotype of each individual because cells in each body may respond differently to different compositions of drugs. For example, people with allergies to Aspirin would have red eyes, hives, lips swelling, etc.[7]

Deep Genomics Inc. and other companies have used its AI program to understand genes and to determine the best medicine to the patient based on their distinctive genotype. These drugs would be assigned to each patient based on his genome, rather than by doctors not knowing thoroughly the potential effects of the drug to the patient.

Furthermore, cancer is caused by the mutation of the DNA[8] and that specific mutation causes the cell to fail to respond to apoptosis. There are a lot of genes that may cause cancer, although mutated cells would usually be killed by the Cytotoxic T cells, cancer cells' Major Histocompatibility Complex (MHC) Protein can fool Cytotoxic T cells so they will be observed as normal[9], and it causes the failure of the body to respond to cancer.

AI can take in the gene from a tumour and analyse the mutation in it causing cancer, and hence provides the best diagnosis to the problem and the best medicine based on the patient's genome, for example, a protein

[4] **Marr, Bernard. 2018.** "How Is AI Used in Healthcare - 5 Powerful Real-World Examples That Show the Latest Advances". Forbes.Com. https://www.forbes.com/sites/bernardmarr/2018/07/27/how-is-ai-used-in-healthcare-5-powerful-real-world-examples-that-show-the-latest-advances/#5e51c05f5dfb.

[5] **"What Is DNA?".** 2019. Genetics Home Reference. https://ghr.nlm.nih.gov/primer/basics/dna.

[6] **"Biological Basis Of Heredity: Recombination And Linkage".** 2019. Palomar.Edu. https://www2.palomar.edu/anthro/biobasis/bio_3.htm.

[7] **"Aspirin Allergy: Know the Symptoms".** 2019. Mayo Clinic. https://www.mayoclinic.org/diseases-conditions/drug-allergy/expert-answers/aspirin-allergy/faq-20058225.

[8] The most commonly mutated genes are TP53, BRCA1, BRCA2 and PTEM.
"Genetics for Cancer". 2017. National Cancer Institute. https://www.cancer.gov/about-cancer/causes-prevention/genetics.

[9] **"How Cancer Cells Escape from Immune System ".** 2016. ScienceDaily. https://www.sciencedaily.com/releases/2016/05/160510094212.htm.

that attaches to the MHC Protein of the cancer cells. These methods can also be used to predict the impact of the mutation on one's body[10] in case if the medicine is taken.

Artificial Intelligence: Surgical Assistance

Numerous surgeries are being carried out daily, and there is a considerable amount of people who die from surgeries every day. Inevitably, all surgeries carry a risk, ranging from infections to death[11]. No matter how high the chance for a risk relating to human biology, the chance of man-made mistakes is not negligible.

AI has been introduced to assist professional surgeons, not only because AI rarely makes a wrong choice, and also, they are not disturbed by emotions, such as fear, love, exhaustion, disgust or stress.

Digital Surgery[12], the first company[13] to develop AI in surgery has aimed to make their products *'support surgical teams to deliver safer surgeries'* by investing 10 million dollars to provide safer surgery to over 5 billion patients[14]. They hope that they can make their flagship product always available free of charge rather than aiming in a profit.

The development of AI in the medical world has raised concern over the medical industry about surgeons would be unemployed as AI could take over their jobs. Although AI would be able to execute the operation better than humans, there are aspects where AI cannot do for a prolonged period of time due to the incapability of understanding and manipulating human emotions[15]. In real life, doctors are capable of understanding human emotions and are able to answer a question to the degree of depth that the patients would want to know. Moreover, doctors are also able to make patients feel more personally cared for.

There are areas around the globe where the medical resource is scarce, for example, in Hong Kong public hospitals, doctors' consultations are limited to 3 minutes and a 30-hour shift is common.[16] The application of artificial intelligence in simple surgeries such as tooth removal would reduce the

[10] **Marr, Bernard. 2018.** "The Wonderful Ways Artificial Intelligence Is Transforming Genomics and Gene Editing". Forbes.Com. https://www.forbes.com/sites/bernardmarr/2018/11/16/the-amazing-ways-artificial-intelligence-is-transforming-genomics-and-gene-editing/#3bd0213042c1.

[11] **"Risks for Surgery".** 2017. National Kidney Foundation. https://www.kidney.org/transplantation/livingdonors/risks-of-surgery.

[12] **"Artificial Intelligence (AI) In Surgery | Digital Surgery".** 2019. Digital Surgery. https://digitalsurgery.com/ai-in-surgery/.

[13] **Armstrong, Reece.** 2018. "First AI Platform for Surgeons Launches - Digital Health Age". Digital Health Age. http://digitalhealthage.com/first-ai-platform-for-surgeons-launches/.

[14] **"Digital Surgery's Investment to Support Access to Safer Surgery".** 2018. Touchsurgery.Com. https://www.touchsurgery.com/blog/unga-safe-surgery/.

[15] **Edwards, Charlotte.** 2018. "Will Automation Take Over Healthcare Jobs? Industry Experts Share Their Opinion - Verdict Medical Devices". Verdict Medical Devices. https://www.medicaldevice-network.com/features/will-automation-take-healthcare-jobs-industry-experts-share-opinion/.

[16] **Tsang, Emily.** 2018. "Hong Kong Public Hospital Crisis". South China Morning Post. https://www.scmp.com/news/hong-kong/health-environment/article/1931049/inside-hong-kongs-public-hospital-crisis-temporary.

stress doctors will have. Hence more saving in time can be allowed which can be used in resting or consultation, therefore they would be less tired and would make better decisions rather than being bombarded by negative emotions.

Revolution to the medical industry

AI has placed an infinite amount of changes to the medical industry but not many of them have actually changed the medical industry that can be categorized as 'revolutionary'.

One definition of AI is that it is an algorithm that has the ability to process complex problems logically and to provide an outcome. It is not a physical property but an algorithm that can be placed in any machines to be executed by the hardware. Hence it would imply that sending the source code of the AI to Developing Countries so they can create their own machinery. This means that people would be able to use cheaper medication. It implies that the cost of medication around the globe would be greatly decreased because the only cost that would be required to be repaid is the manufacturing cost, and it is not expensive.

In general, new medical treatments and medicine can be quickly synchronized in terms of hardware improvement without needing doctors to listen to lectures and talks about the most updated medical information, and potentially getting a license in order to use the technology. Furthermore, since data is synchronized, ML results can be shared very quickly across the globe with all the machines, such that they can improve all the time. All these would mean that the flexibility and the speed of learning what to do or what not to do can be rapid and far-reaching.

Research shows that healthcare data has reached 153 Exabytes in 2013 and is predicted to reach 2,314 Exabytes by 2020[17]. It means that it would take 2,426,404,864 1TB hard-disks to store the data by 2020. Assuming that each hard-disk is 5cm thick and is stacked. It would be able to travel from north pole to south pole 9 times. Such a tremendous amount of data is not possible to be read by one single human within a few years to understand, whereas AI is able to record all knowledge in a single database and to retrieve the correct piece of data whenever it is required.

Furthermore, personalized medicine can be cheaply produced. Rather than eating Paracetamol which is the 'one for all' medicine, AI can create medicine that is the best fit to the person in need and its efficiency and dosage can be controlled by the AI based on your genome and physical characteristics. It signifies that the sickness can be taken care of easily and properly.

Despite all the advantages that AI can bring, there are a lot of worries about personal security because their source code is too complicated to understand and is not open-sourced. People without medical or computational knowledge are not convinced that a

[17] **"Medical Data Storage".** 2018. Healthdataarchiver.Com. https://www.healthdataarchiver.com/health-data-volumes-skyrocket-legacy-data-archives-rise-hie/.

computer could achieve the same competency as a student undertaking year of study in order to be licensed as a doctor. Furthermore, no one can guarantee that the algorithms in the AI are always correct in all circumstances. The outcome of a faulty algorithm is disastrous, and the worst-case scenario is leading to the death of the patient not because of their health or disease but because of technical failure.

Since machines are connected to the internet in order to synchronize, the genetic scanning result would be available online. It leads to a problem with personal privacy since you will never know who will read your data. Furthermore, this data will be vulnerable to be hacked.

The inability of trusting AI is a large setback to revolutionize the medical industry. Even, if doctors and AI would be equally professional, it feels warmer and more comfortable talking to a person rather than communicating to an AI. The concern about health and own personal life is equally important. A natural instinct for a human is to trust a doctor rather than a machine.

In the real application, AI plays an important role when doctors have to come to a major decision because the AI has the ability to diagnosis the symptom for the right medicine for the patient faster and more accurately. However, AI is unable to replace doctors because of their inability to provide psychological comfort to the patients and trust to the family members.

Without AI, doctors would not be able to have access to the entire world's library of diseases and AI can allow a doctor to widen his analytical horizon to think more about something that he would not usually think of.

AI has played a large role in the medical industry. It has revolutionized the medical industry by providing alternative pathways to problems and potentially making a solution that professionals would not have thought of or need to think of. However, since they are not competent to create a psychological bond with the patient, which causes a lack of trust between the client and the patient, it hinders a further development of AI in the modern world to revolutionize the medical advancement.

Conclusion

AI has been developed at a very fast pace since the invention of computers and the rapid development of speed and simultaneous connection. Different engineers have contributed their knowledge about AI to build a better place for humans to live, including the development of AI for the medical industry,

There are numerous applications of AI for the medical world, most commonly known as drug manufacturing or disease diagnosing. Mobile apps such as Ada have been developed to diagnose diseases based on the symptoms.

AI can be used to analyse one's genome to create personalized drugs at a cheaper cost which can be

more effective to a patient with a relatively low chance of allergenic reactions compared to the traditional 'one for all' medicine.

Furthermore, AI has been in place to work with surgeons such that surgeons can have more flexibility in time, widened their mindset on what the disease is and the ways to perform a surgery with a higher chance of success.

The collaboration in time and knowledge sharing between AI and humans can greatly allow the medical industry to advance. However, patients are usually reluctant to trust a machine because of various reasons, including incorrect algorithms and psychological effects. Such a feeling has hindered AI to revolutionize the medical industry further nowadays.

As a conclusion, AI has aided humans a lot in the medical world, but more efforts need to be contributed so AI can advance even further with an ultimate aim in making AI a faithful partner to patients to gain their trust on the machines.

Summarization of Document

Abbreviations Used:

AI	Artificial Intelligence
MIT	Massachusetts Institute of Technique
CMU	Carnegie Mellon University
ML	Machine Learning
DNA	Deoxyribonucleic Acid
MHC	Major Histocompatibility Complex

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References

Artificial Intelligence (AI) In Surgery | Digital Surgery. 2019. Digital Surgery. https://digitalsurgery.com/ai-in-surgery/.

Aspirin Allergy: Know the Symptoms. 2019. Mayo Clinic. https://www.mayoclinic.org/diseases-conditions/drug-allergy/expert-answers/aspirin-allergy/faq-20058225.

Biological Basis of Heredity: Recombination And Linkage. 2019. Palomar.Edu. https://www2.palomar.edu/anthro/biobasis/bio_3.htm.

Digital Surgery's Investment to Support Access to Safer Surgery. 2018. Touchsurgery.Com. https://www.touchsurgery.com/blog/unga-safe-surgery/.

Genetics for Cancer. 2017. National Cancer Institute. https://www.cancer.gov/about-cancer/causes-prevention/genetics.

How Cancer Cells Escape from Immune System. 2016. ScienceDaily. https://www.sciencedaily.com/releases/2016/05/160510094212.htm.

Medical Data Storage. 2018. Healthdataarchiver.Com. https://www.healthdataarchiver.com/health-data-volumes-skyrocket-legacy-data-archives-rise-hie/.

Most Common Drugs That Cause Allergies. 2019. WebMD. https://www.webmd.com/allergies/most-common-drugs-that-cause-allergies.

Risks for Surgery. 2017. National Kidney Foundation. https://www.kidney.org/transplantation/livingdonors/risks-of-surgery.

What Is DNA?. 2019. Genetics Home Reference. https://ghr.nlm.nih.gov/primer/basics/dna.

World-Information.Org. 2019. World-Information.Org. http://world-information.org/wio/infostructure/100437611663/100438659360.

Armstrong, Reece. 2018. First AI Platform for Surgeons Launches - Digital Health Age. Digital Health Age. http://digitalhealthage.com/first-ai-platform-for-surgeons-launches/.

Edwards, Charlotte. 2018. Will Automation Take Over Healthcare Jobs? Industry Experts Share Their Opinion - Verdict Medical Devices. Verdict Medical Devices. https://www.medicaldevice-network.com/features/will-automation-take-healthcare-jobs-industry-experts-share-opinion/.

Marr, Bernard. 2018. How Is AI Used in Healthcare - 5 Powerful Real-World Examples That Show the Latest Advances. Forbes.Com. https://www.forbes.com/sites/bernardmarr/2018/07/27/how-is-ai-used-in-healthcare-5-powerful-real-world-examples-that-show-the-latest-advances/#5e51c05f5dfb.

Marr, Bernard. 2018. The Wonderful Ways Artificial Intelligence Is Transforming Genomics and Gene Editing. Forbes.Com. https://www.forbes.com/sites/bernardmarr/2018/11/16/the-amazing-ways-artificial-intelligence-is-transforming-genomics-and-gene-editing/#3bd0213042c1.

Tsang, Emily. 2018. Hong Kong Public Hospital Crisis. South China Morning Post. https://www.scmp.com/news/hong-kong/health-environment/article/1931049/inside-hong-kongs-public-hospital-crisis-temporary.

Vyacheslav, Polonski. 2018. The Conversation. http://theconversation.com/people-dont-trust-ai-heres-how-we-can-change-that-87129